Verified fact record — product spec

One verified, versioned set of facts per property unit, so a voice agent can answer buyers and renters from current, cited information only. Written for Jigar at the Grok Bot Real Estate Hackathon, London, 12 November 2026. Code freeze is 23:00 Europe/London.

Contents

Product and customer
Stakeholder and data-source map
Ingestion pipeline
Core features
Contradiction handling
Fact catalogue
Voice agent contract
Evidence from tonight
Demo, roadmap, risks and next test

How to read this page

Evidence is from tonight's hackathon material (live listings, 185 sale brochures, Land Registry Price Paid matching, and Dwelly's synthetic cases) or from a document we opened. Assumptions are labelled as such. No market-size, frequency, or willingness-to-pay figures are invented. Wrong or stale material facts in marketing can breach UK consumer-protection rules and the National Trading Standards Estate and Letting Agency Team (NTSELAT) material information guidance that estate and letting agents are expected to follow. That duty is mentioned only in general terms. This page is not legal advice and does not cite clause numbers.

Product and customer

Property facts start in many places — the developer, the architect, the sales agency, the letting agent, the conveyancer, the managing agent, and official registers such as Land Registry and the EPC register — and they change as a development moves from planning to sale to management. The same figure is copied from document to document, often by a different team, and the copies disagree: one brochure says

showers, another says baths; one set of particulars says £700,000, a later one says £600,000. This product ingests every available source for a development and for each unit, extracts structured facts with their provenance (document, page, quoted text, date, and author or owner), detects contradictions, applies resolution rules, and requires a human agent to verify anything the rules cannot settle. The result is one versioned fact record per unit. A voice agent — the customer interface for buyers and renters — answers only from that current verified record, cites the source and date, flags anything unverified or still in conflict, and escalates rather than guessing, so a caller is not left acting on the wrong information.

Paying customer

Assumed to be the sales or letting agency, or the developer that instructs them. They own the documents, they are responsible for material facts in marketing, and they take the enquiry. We have not tested who would buy this or what they would pay. That is an assumption, not a finding from tonight's data.

End user

The buyer or renter who calls. They never see the workbench. They hear only verified current facts, or a clear refusal and an escalation to a human agent, with their details taken as a lead.

The product is not a general property-search wrapper, and it is not a chatbot over raw PDFs. The voice layer is deliberately thin. Its job is to speak from a checked record, not to invent a search experience the judges will have already seen.

Stakeholder and data-source map

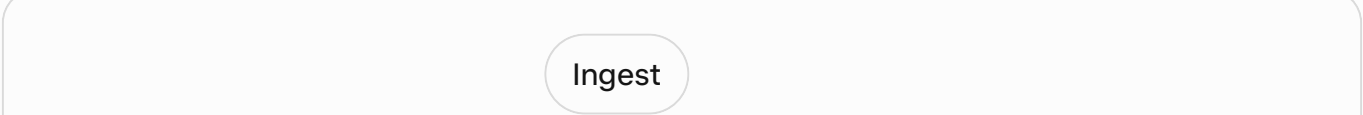
In a real sales and lettings process, facts pass through several teams and document types, not just brochures. Tonight's bulk Drive folder is sales brochures, so the other document types below are the product's intended sources — and the roadmap — not what we can demo from that folder. The lifecycle runs: land and planning, design and specification, pre-launch pricing, marketing, enquiry and viewing, reservation and offer, conveyancing, completion and handover, then letting and management.

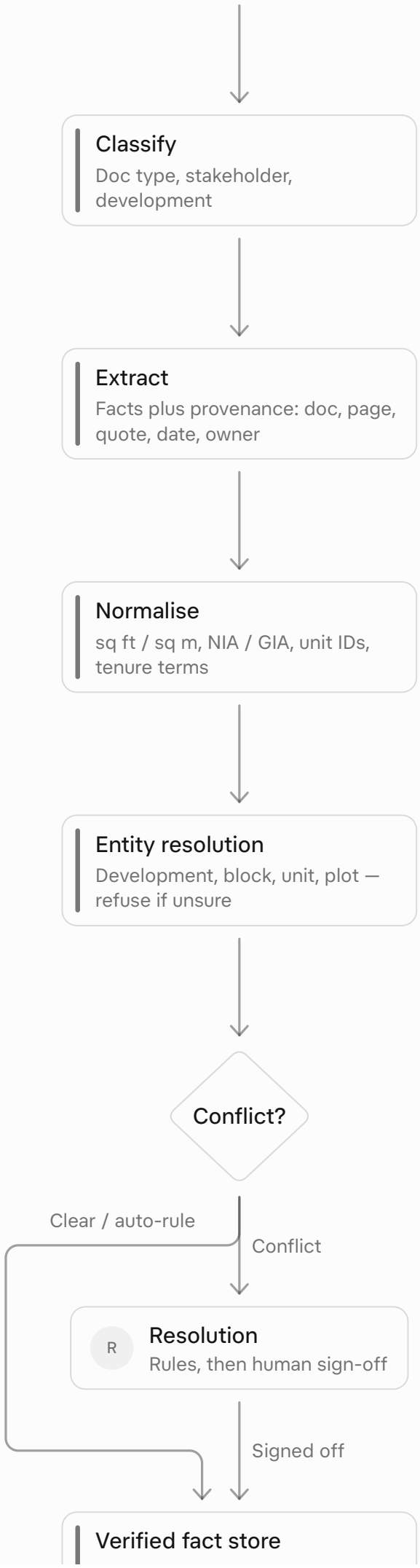
Stakeholder	Documents and data they produce
Developer / property owner	Specifications, floor plans, schedules of accom lists, unit allocations, marketing briefs, handove
Architects and planning consultants	Planning application, drawings, Design and Acc specifications, planning conditions, as-built dra
Sales and marketing agency	Sales brochures, formal particulars, CGIs, webs
Letting and sales agents (front line)	Portal listings, CRM notes, availability, viewing f marketing
Conveyancers and solicitors	Draft contracts, title packs, searches, formal er report on title
Managing agents	Service-charge budgets, residents' handbooks, maintenance quotes
Land Registry	Title register and plan, tenure, proprietorship, P
EPC register	Energy rating, certificate floor area, heating fue
Warranty and building-safety assessors	NHBC or equivalent warranty, EWS1 or success fire-risk and building-safety documents
Buyer or renter	Enquiries only — they consume facts, they do r

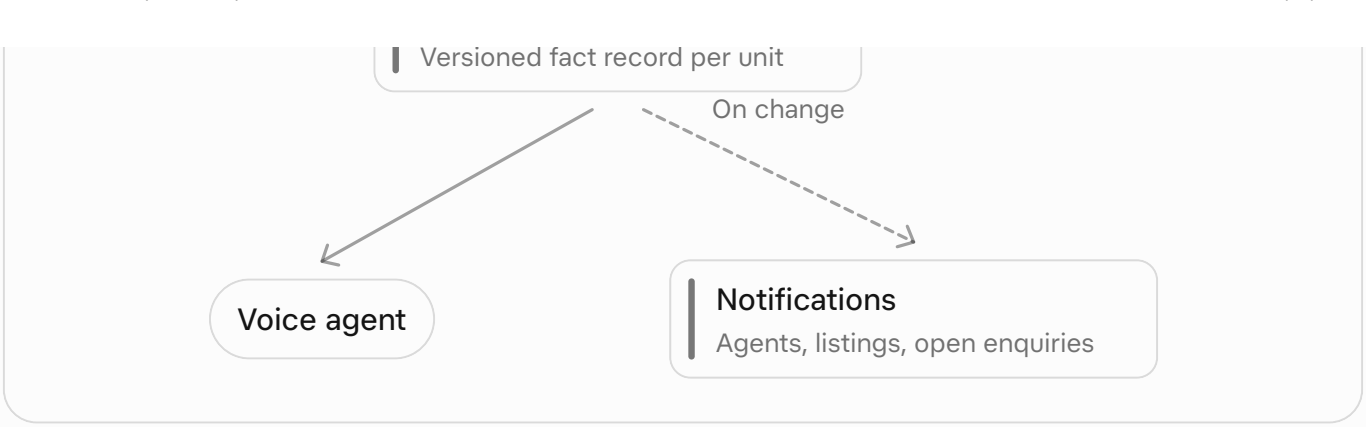
Update frequency and raw trust are typical values (assumption), except where tonight's data gives a concrete example. Raw trust is how much an unverified document from this source should be believed before resolution.

Ingestion pipeline

Every source, from every stakeholder above, enters the same pipeline and lands in the same fact record. A fact is not visible to the voice agent until it is verified, or until a resolution rule has settled it and a human has not been required to intervene.




Classify
Doc type, stakeholder, development
Extract
Facts plus provenance: doc, page, quote, date, owner
Normalise
sq ft / sq m, NIA / GIA, unit IDs, tenure terms
Entity resolution
Development, block, unit, plot — refuse if unsure
Conflict?
Clear / auto-rule
Conflict
R
Resolution
Rules, then human sign-off
Signed off
Verified fact store



Entity resolution is the step that makes every later comparison trustworthy. A join is when the system links a listing or a document to an official record, or two documents to each other, because they look like the same property. A bad join compares the wrong properties and raises a false conflict. Tonight's evidence: organiser pre-joins in the sample were often wrong (Flat 53 Coachman Court was matched to Flat 21), and a first pass of tenure mismatches was mostly bad joins, not real errors. The rule is strict: require a postcode and a house number or building name; do not merge on postcode alone; reject an agent's office address used in place of the property; do not use organiser pre-joins as ground truth; if the match is not confident, refuse and hold the item unmatched. Do not invent a conflict from an uncertain join.

Normalisation is what stops measurement artefacts looking like disputes. Store both square metres and square feet, and always label the basis (NIA, GIA, or another stated standard). Keep unit, plot and type identifiers as extracted, and map them only after resolution. Treat freehold, leasehold and share of freehold as a controlled vocabulary, not free text.

Change propagation

1. **A new or revised source arrives**
   It is ingested through the same pipeline, whether it is a new brochure, a listing edit, or an official-record refresh.
2. **It is attached to the right unit, or held**
   If entity resolution is not confident, the source is unmatched and no live fact changes.
3. **New values are compared with the current fact record**
   Each field is classified: identical, supersession, measurement or scope difference, or true conflict.
4. **Supersession updates the record**
   A later document of the same type from the same issuer can replace the value. The previous value stays in history, with who published it and when.
5. **A conflict takes the fact off the voice channel**

The field becomes pending or conflicting. The voice agent stops quoting it. The human agent is notified to sign it off.

6 **After sign-off, the new verified value is published**

Listings that still show the old value are flagged. Open enquiries that were told a previous value are queued for a correction — a later feature, not tonight's demo.

Core features

MVP is what the first usable product, and tonight's demo, must do. Later is the roadmap once the core loop works.

MVP	Multi-source ingest · MVP Take in PDF particulars, listing rows, and Land Registry Price Paid records where a match is confident. Without more than one source, there is nothing to reconcile.	Build tonight
MVP	Classification · MVP Identify document type, stakeholder, development and date, so later rules know what they are comparing.	Build tonight
MVP	Extraction with full provenance · MVP Every structured fact carries the document, page, quoted text, date, and author or owner. Traceable data is scored on every hackathon track, and the voice agent cannot cite a source it does not have.	Build tonight
MVP	Normalisation · MVP Currency, area in both sq m and sq ft, NIA vs GIA labelled, tenure vocabulary, and unit identifiers. This is what revealed the Kelso Villa floor-area error as a conversion and basis mistake, not two different buildings.	Build tonight
MVP	Strict entity resolution · MVP Match development, block, unit and plot with a refuse-to-guess rule. This is the real engineering. A sloppy matcher cries wolf, and a judge who clicks one false flag stops trusting the rest.	Build tonight
MVP	Conflict detection and taxonomy · MVP Classify each disagreement (true conflict, supersession, measurement, scope, plot or type, stale listing, official-record mismatch) so the right rule, and the right voice script, can run.	Build tonight
MVP	Resolution workbench · MVP Suggest a rule-based outcome, then let the human agent accept or override it. Record who chose, when, and a short reason. Without this, there is no verified record and no audit trail.	Build tonight

there is no verified record and no audit trail.

MVP	**Versioned fact record per unit · MVP** One current value per field, with full history. This is the object the voice agent reads. It is also what makes a later correction explainable.	Build tonight
MVP	**Official-record check · MVP** Where a Land Registry match is confident, compare tenure and property type and flag mismatches as 'check this', not as automatic errors. Complements brochure-to-brochure reconciliation and is stronger on provenance.	Build tonight
MVP	**Voice agent, thin and strict · MVP** The customer interface. Answers only from verified current facts, cites source and date, uses the four confidence states, and escalates with lead capture rather than guessing. One working query is enough for tonight; polish is the first thing to cut if time runs short.	Build tonight, keep thin
Later	**Stakeholder-system connectors · Later** Developer document rooms, agent CRMs, portal feeds, managing-agent systems. The business only works if facts can be read where they already live. Access to those systems is a stated kill risk, not a solved problem.	Roadmap
Later	**EPC register and live title · Later** Energy rating and certificate floor area, and current title rather than Price Paid at last sale. The EPC register needs a GOV.UK sign-in and is not in the event Drive, so it is out of tonight's demo.	Roadmap
Later	**The rest of the document types · Later** Leases, contracts, specifications, price lists, planning documents. Tonight's bulk folder is sales brochures. These types belong in the product; they are not what we can load for the demo.	Roadmap
Later	**One development, start to finish · Later** A single scheme followed from planning through management. Tonight's worked examples are separate properties, and should be presented as such.	Roadmap
Later	**Notifications and listing updates · Later** Tell the agent, and eventually the portal row, when a verified fact changes, and correct callers who were given a previous value.	Roadmap
Later	**Compliance pack · Later** An export of current verified facts, sources, and sign-offs, for material-information due diligence. Mentioned generally; not designed as a legal product tonight.	Roadmap

Later Richer voice · Later Roadmap

Qualification, viewing booking, and multi-unit comparison — still verified-only. Do not build this until the fact record is solid.

Contradiction handling

Not every disagreement is a true conflict. The product classifies first, then applies a rule, then asks a human where the rule is not enough. The voice agent does not quote a field that is conflicting or pending.

Source-authority hierarchy

Default order, highest first. Recency applies within the same tier. A newer brochure does not beat a contract. A human must sign off true conflicts, official-record mismatches, and any override of a higher-authority source. The hierarchy is a product rule (assumption about what should win), informed by how these documents are used, not a finding from tonight's data.

1. Signed legal instrument: current title, lease, sale contract, reservation agreement.
    - Per field: tenure, lease length, ground rent, covenants, pets and subletting, and the contracted price sit here once signed.
2. Current official register extract: Land Registry title, EPC certificate — always with the register's date.
    - Caveat: a Price Paid tenure is as at the sale date and may lag a later private transaction (Lumley House). An EPC floor area has its own measurement basis.
3. Latest signed specification or measured survey, with the basis stated (NIA, GIA, or other).
    - Floor areas, dimensions, specification, finishes, bathrooms, accessibility, heating and solar as designed or as built.
4. Latest signed price list or formal developer instruction.
    - Asking price and incentives, before a contract exists.
5. Latest marketing brochure or formal particulars.
    - Kelso Villa: the October 2026 Carter Jonas particulars sit above the April 2026 particulars for the same issuer and document type.
6. Portal listing, CRM note, or verbal remark.
    - Lowest tier. Useful as a prompt to check, not as a value to quote. Kelso listings

still at £700,000 are stale against the October brochure.

Type	What it is
True conflict	Same unit, same field, same measurement basis, va cannot both be right
Version supersession	A later document from the same issuer replaces an
Unit or measurement mismatch	The same quantity expressed in different units or o basis; both may be right, or the conversion may be
Scope difference	A development-wide claim versus a per-unit fact
Plot or type variation	Different plots or house types have different specif
Stale listing versus latest document	A portal or CRM row was not updated after a new b price list
Official-record mismatch	Marketing or a listing disagrees with Land Registry register

> **Not a contradiction: a bad join**
> If entity resolution is not confident, do not raise a conflict. Flat 53 Coachman Court matched to Flat 21 is a matching failure, not a dispute about Flat 53. Treating it as a fact conflict would train both agents and judges to distrust the product.

Fact catalogue

Category	Typical authoritative source	Why it drifts	Risk if wrong
Price and incentives	Latest signed price list or developer	Price cuts and incentives are issued	A buyer acts on the wrong price. Asking

incentives	list or developer instruction; reservation agreement once a unit is under offer	incentives are issued in a new list or brochure and not pushed to the portal. Evidence: Kelso Villa April £700,000 vs October £600,000; listings still at £700,000	wrong price. Asking price is material information in marketing
Tenure, lease length, ground rent, service charge	Current Land Registry title and the lease; service charge from the lease and management accounts	Marketing shorthand; stale Price Paid tenure after a later transaction; listing rows that disagree with each other. Evidence: Kelso tenure rows; Lumley House listing vs 2016 Price Paid	Wrong tenure or charges are serious material-information failures and a common source of complaint
Floor areas and dimensions	Measured survey with the basis stated (NIA, GIA, or other); EPC floor area is official but on its own basis	Square-metre to square-foot conversion errors; NIA stated as GIA. Evidence: Kelso April brochure printed 185 sq m as 2,700 sq ft	The buyer overestimates space; pounds-per-square-foot and comparables are wrong. Area is material information
Specification, finishes, bathrooms	Latest signed specification or schedule of finishes, at unit or plot level	Brochure versions, show-home versus actual unit, plot-type variation. Teammate's shower vs bath is one person's account, not tonight's documents	Expectation gap, wasted viewings, complaints after reservation
Energy: EPC, heating, solar	Official EPC register for the rating and certificate date; specification or as-built pack for heating and solar	Predicted versus as-built ratings; marketing claims ahead of a certificate. EPC is not in tonight's demo (GOV.UK sign-	Energy running-cost expectations and material information on the rating

		in)	
Parking and EV charging	Lease, title or deed of grant, and the allocation schedule	Spaces reallocated; 'parking available' versus a named space; EV charging promised then delayed (assumption, typical)	A frequent deal-breaker. Wrong allocation is a complaint and may be material
Completion dates and build stage	The sale contract once exchanged; before that, the developer's current programme, not a marketing 'ready by'	Marketing dates slip; listings are not updated (assumption, typical)	Chains break, bridging costs arise, a move for school or work is missed
Warranties (NHBC or equivalent)	Warranty-provider documentation and the build certificate	Cover is assumed as standard, or quoted before it is in place, especially on conversions (assumption, typical)	The buyer believes they are protected when they are not
Building safety, cladding, EWS1	EWS1 or successor assessment, building-safety case, landlord or managing-agent disclosure	Assessments are updated or buildings remediated; marketing stays silent (assumption, typical). Not in tonight's data	Mortgageability, safety, and material information. Do not let a voice agent guess here
Broadband	A provider or Openreach confirmation, or a clearly labelled prediction	A development-wide 'full fibre' claim versus the unit's actual connection (assumption, typical)	Smaller than tenure or price, but a common enquiry. A wrong answer wastes a viewing
Restrictions and covenants	Title register, lease, and planning conditions	Omitted from marketing; a general development claim versus a unit-specific restriction (assumption, typical)	The buyer cannot use the property as intended. This is material information
Pets and subletting	The lease and any landlord or management policy	Verbal assurances from an agent versus the lease wording	A renter or buyer cannot keep a pet or sublet. A common

		(assumption, typical)	post-move complaint
Availability and status	The live reservations system or the agent's current stock list	Portal lag; reserved units still advertised (assumption, typical)	Wasted enquiry, a poor experience, and missed genuine leads
Accessibility	Measured drawings, access statement, as-built survey	CGI and brochure claims versus actual step-free access, lift provision, or door widths (assumption, typical)	Material for disabled buyers and renters; wasted viewings; discrimination risk
Management company	The lease, the handover pack, and Companies House for the company itself	Name changes; interim developer management versus the long-term agent (assumption, typical)	The buyer does not know who to pay or which standards apply

Technical facts that are prone to drift as they are copied along the modern sales flow: planning into specification, into the price list, into the brochure, into the portal listing, into what is said on the phone. Each handoff is a copy, often by a different team, often without the measurement basis or the date.

Voice agent contract

The voice agent is the customer interface, not the product's centre of gravity. It may only speak from the verified fact record. These rules are the contract between the fact store and the caller.

1. Answer only from the verified current fact record for a confidently matched unit.
2. Cite the source and its date on every factual answer.
3. State the confidence: verified, pending, conflicting, or unknown.
4. If the state is pending, conflicting, or unknown: do not quote a number or a yes-or-no as fact; explain briefly; escalate to a human agent; capture the caller's name and number as a lead.
5. If the unit is not confidently matched, treat the question as unknown. Never answer from a neighbouring unit.
6. If a listing disagrees with the verified record, quote the verified record and, if it helps the caller, say the listing looks out of date.

7. Do not fill gaps from general knowledge, from other developments, or from the model's own priors.
8. After a fact is updated, later answers use the new version. Proactive correction of earlier callers is a later feature.

Confidence	Meaning	What the caller hears
Verified	A current value has been signed off, or a rule has settled it and human review was not required	The fact, the source, and the date. Older superseded figures only if useful, and clearly labelled as older
Pending	A value has been extracted but not yet verified, and there is no active conflict	A figure is in the file but has not been verified, so it will not be quoted. Offer to have the agent confirm and take a lead
Conflicting	Two or more sources disagree and a human has not signed off	Do not quote either value as fact. Say that sources disagree, that an agent will confirm, and take a lead
Unknown	No source, or no confident unit match	We do not have a verified answer and will not guess. Offer to ask the agent and take a lead

Example: verified price, Kelso Villa

> What's the asking price for Kelso Villa?
> Buyer

> The current asking price is £600,000, on a freehold-offers basis, from Carter Jonas's October 2026 particulars. That figure was verified by the agent on [date]. An earlier April 2026 brochure said £700,000, and the portal listings we have still show £700,000. Those are out of date against the October particulars.
> Voice agent, verified

Example: conflicting price, not yet signed off

> What's the asking price for Kelso Villa?

> What's the asking price for Kelso Villa?
>
> Buyer

> I can't give you a verified price yet. Two sets of particulars disagree: April 2026 said £700,000 and October 2026 said £600,000. I won't guess. I'll pass this to the agent to confirm and come back to you. Can I take your name and a number?
>
> Voice agent, conflicting

Example: floor area, after verification

> How big is it?
>
> Buyer

> The net internal area is 185 square metres, about 1,990 square feet, from the October 2026 particulars, verified on [date]. The gross internal area is 2,700 square feet. An April brochure printed 185 square metres as 2,700 square feet, which is a conversion error — 2,700 is the gross figure. I can send both figures with the basis labelled.
>
> Voice agent, verified

Example: official-record mismatch, Lumley House

> Is Lumley House freehold?
>
> Buyer

> I can't confirm tenure yet. The listing says freehold, but the Land Registry Price Paid record for the 2016 sale is leasehold. That official record may be out of date if the freehold was bought later, so this needs a check, not a yes or no from me. I'll ask the agent to confirm against the current title and call you back. Can I take a name and number?
>
> Voice agent, conflicting

Example: unknown

> What's the EPC rating?
>
> Buyer

> I don't have a verified energy rating for this property, and I won't guess. I can ask the agent to look it up and get back to you. Can I take your name and a number?
>
> Voice agent, unknown

Voice agent, unknown